

PHASE-A RESULTS: THE FREE EMA-KL SURROGATE PREVENTS STALENESS COLLAPSE, AND ITS THEORY PICKS THE RIGHT DOSE

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ABSTRACT. Asynchronous RL trains on rollouts from stale policy snapshots; the free EMA-KL surrogate $c(\log T_n - \log T_{n-\Delta})$ with $c = \alpha/(\Delta(1 - \alpha))$ regularizes against an implicit EMA anchor at zero extra cost. On Qwen3-4B/Countdown (TBA, no centering) we find: (1) **without** the penalty, training *collapses* at high staleness — homogeneous $\Delta=32$ falls from eval 0.75 to 0.14 by step 200, and a heterogeneous mixture $\Delta_i \in \{1, 4, 10, 32\}$ collapses ~ 60 steps later ($0.75 \rightarrow 0.002$ by step 300); (2) **with** the penalty at the theory-prescribed strength, every arm is stable and reaches eval 0.79–0.83; (3) per-rollout calibration $c_i = \alpha/(\Delta_i(1 - \alpha))$ matches the *best* uniform dose found by a $3\times$ sweep (peaks 0.816 vs. 0.817) — calibration’s value is that the theory selects the right strength *a priori*, without a sweep; the $3\times$ larger uniform dose costs ~ 1.5 points. A homogeneous $\Delta=10$ reference through the new machinery reproduces the historical baseline (0.829 vs. 0.827–0.831), and per- Δ telemetry confirms the surrogate’s predicted small- Δ variance (rel. error 5.9 at $\Delta \leq 2$ vs. 0.19 at $\Delta \geq 21$) without harming training. (4) Under symmetric NVFP4 emulation, quantization *compounds* with staleness: the FP4 $\Delta=32$, $\beta=0$ arm collapses ~ 70 steps earlier than its BF16 twin with a lower peak, while $\Delta \lesssim 10$ stays stable; the *same* theory dose $\beta c = 0.0045$ keeps $\Delta=10$ healthy and extends the $\Delta=32$ collapse boundary $\sim 3\times$ (though not to full-horizon stability — dose scaling for quantization is untested). An asymmetric arm (BF16 trainer, FP4 sampler) fails informatively: the un-centered quantization gap in the advantage-level penalty acts as a global negative reward. (5) In the original-hypothesis cell — FP4 with *heterogeneous* staleness $\Delta_i \in \{1, 4, 10, 32\}$ — the $\beta=0$ control collapses at $\sim$ step 205 while both KL arms train healthily through their full 24 h horizons; on eval, per-rollout staleness-weighted and uniform dosing *tie* (peaks 0.771 vs. 0.764, finals 0.771 vs. 0.764, single seed): the hypothesis’s rescue claim holds, but the weighting refinement is not separable from a well-chosen uniform dose here.

1. SETUP

All runs: Qwen3-4B-Instruct-2507 on Countdown with TBA (no centering, importance sampling on), lr 10^{-6} , $K=16$, seq. 4096, 720 target steps, eval every 20 steps, $\alpha = 0.9$, 2 nodes ($4\times H100$ inference / $4\times H100$ training); 24 h allocations (curves end at steps 490–690). Configs in Appendix A.

Heterogeneous staleness. `--rl.staleness-offsets` $o_{0..3}$ pins each of 4 data-parallel vLLM ranks to a fixed checkpoint lag (reload every step to step $-o_r$). K -groups stay policy-homogeneous; training batches pool ranks and therefore mix policy ages (verified in the rollout parquets: the trainer’s step-10 batch carries generating checkpoints $\{9, 6, 0, 0\}$). The homogeneous arms use offsets 10, 10, 10, 10 (reference) or the legacy reload rule at `async_level=32` ($\beta=0$ control).

Anchors and calibration. Every arm anchors both the importance ratio and the KL term on the sampler’s own vLLM logprobs. Per-rollout calibration uses each sample’s true lag: $c_i = \alpha/(\Delta_i(1 - \alpha))$, $\Delta_i \geq 1$; the uniform arms apply one dose $w = \beta c_{\text{glob}}$ to every rollout ($c_{\text{glob}} = 0.9/3.2 = 0.28125$). The per-token clamp is in raw log-ratio units (± 10 before the coefficient), so truncation is Δ -invariant.

2. RESULTS

Finding 1: staleness alone collapses training; the free surrogate prevents it. The homogeneous $\Delta=32$, $\beta=0$ control peaks at eval 0.752 (step 100), then falls off a cliff between steps 160–180 (train reward $0.68 \rightarrow 0.19$; eval 0.14 at step 200) and oscillates around 0.2–0.45 for 400 more

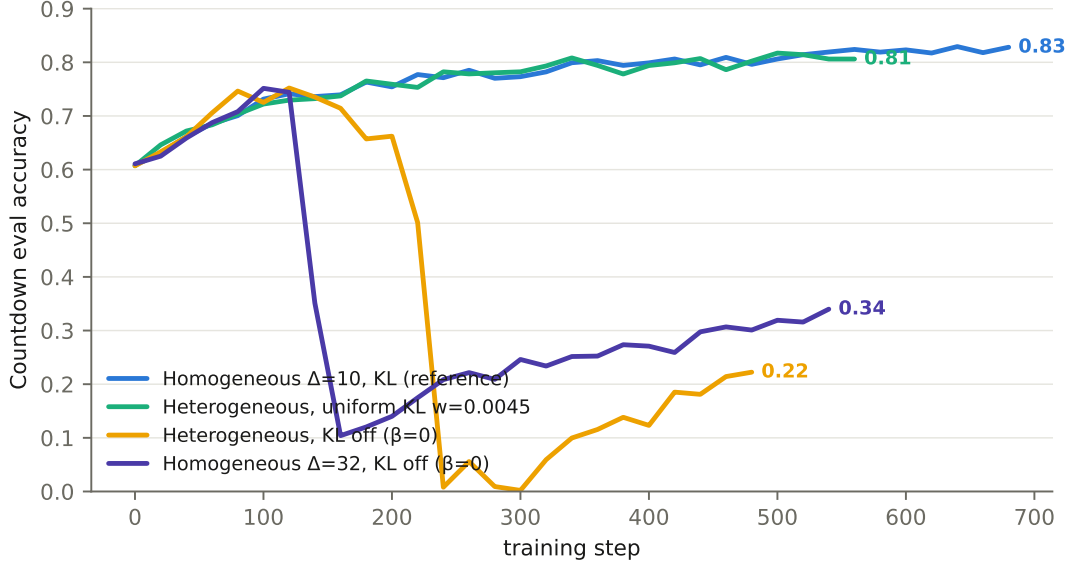


FIGURE 1. Collapse without the KL penalty; stability with it. Yellow (heterogeneous) and violet (homogeneous $\Delta=32$) are $\beta=0$: both collapse (steps ~ 230 and ~ 170) and never recover. Aqua (uniform dose $w=0.0045$) trains through the same window untouched; blue is the homogeneous $\Delta=10$ reference.

steps without recovering. The heterogeneous $\beta=0$ control — average lag $4\times$ smaller — shows the same failure delayed by ~ 60 steps, preceded by a characteristic train-reward overshoot (0.94 at step 127), then eval 0.002 by step 300. Every arm with the EMA-KL surrogate on, at any tested strength, trains stably through and far beyond these windows. This is BF16 end to end: staleness alone, no quantization, breaks unregularized async RL at these lags, and the zero-cost surrogate fixes it.

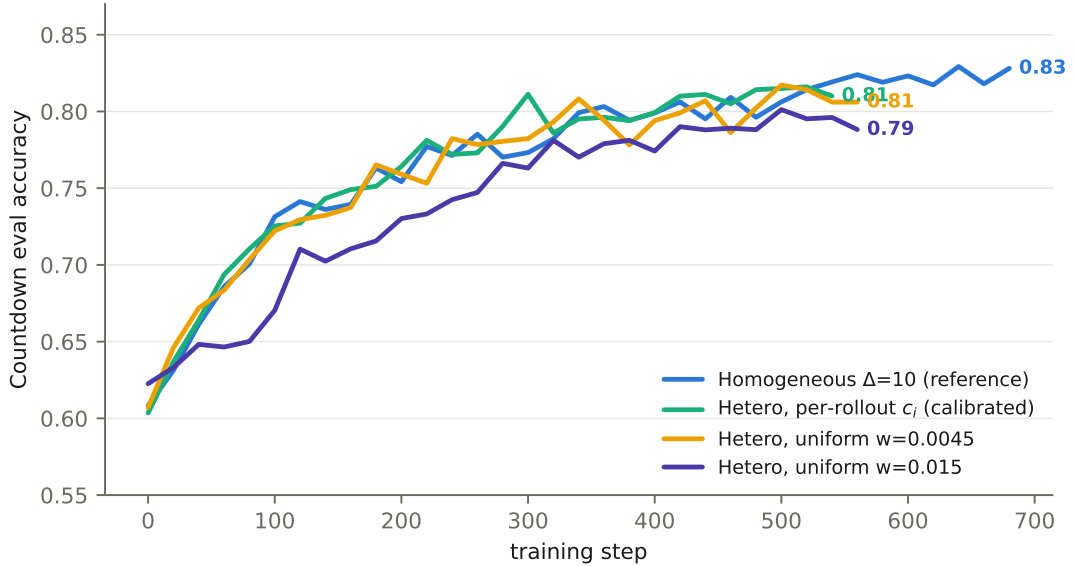


FIGURE 2. All stable arms. Calibrated per-rollout c_i (aqua) vs. the two uniform doses (yellow $w=0.0045$, violet $w=0.015$) under the same $\Delta_i \in \{1, 4, 10, 32\}$ mixture; blue is the homogeneous reference.

Finding 2: calibration matches the best swept dose — without the sweep. Peaks: calibrated 0.816 (step 520), uniform $w=0.0045$ 0.817 (step 500), uniform $w=0.015$ 0.801; reference 0.829. The calibrated arm and the best uniform arm are statistically indistinguishable at one seed, and both sit ~ 1 point under the homogeneous reference. Two honest readings. First, on this task and Δ spread, per-rollout weighting adds no *performance* over the best *tuned* uniform dose — the group-summed penalty appears to care mostly about total strength. Second, the best uniform dose is exactly the one the theory prescribes ($w = \beta_{\text{ref}} c_{\text{ref}} = 0.0045$, transported from the known-good homogeneous run via the βc invariance), and tripling it costs 1.5 points: the calibration theory’s practical payoff here is *dose selection without a sweep*, which is also what a practitioner changing Δ (or mixing Δ_i) actually needs.

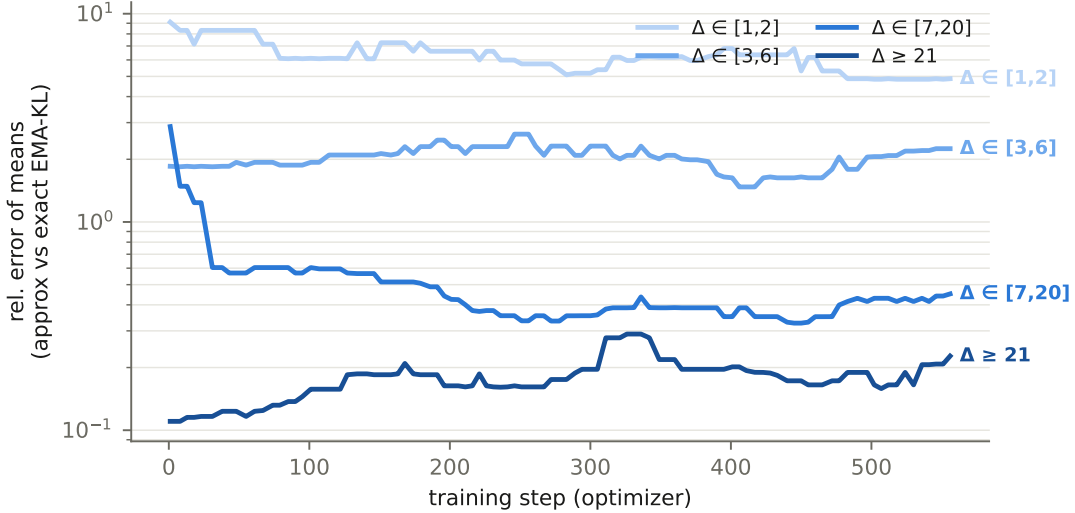


FIGURE 3. Relative error of the surrogate vs. exact EMA-KL by lag stratum (calibrated run; window-median smoothed; log scale).

Finding 3: the surrogate’s error is strongly Δ -dependent — and training doesn’t care. Median rel. error of means: 5.88 ($\Delta \leq 2$), 2.13 ($[3, 6]$), 0.48 ($[7, 20]$), 0.19 (≥ 21) — the theory note’s finite-difference variance warning, measured. Despite a quarter of tokens carrying rel.-error- ~ 6 estimates, the calibrated arm matched the reference; the summed, β -scaled penalty is robust to zero-mean estimator noise. Suggested refinement (untested): floor Δ_i at ~ 4 in c_i .

Finding 4: mechanism and baseline controls. The homogeneous $\Delta=10$ reference through the full new machinery (offsets mechanism, vLLM anchor, raw-units clamp) peaks 0.829 / final 0.828 — inside the historical no-centering band (0.827–0.831).

3. FP4 RESULTS

All FP4 arms use symmetric NVFP4 QDQ emulation (STE on trainer forwards, QDQ on sampler checkpoints, bit-exact between the two) unless noted; same Qwen3-4B/Countdown setup as above.

Finding 5: quantization compounds with staleness, and the same dose rescues it. The $\beta=0$ ladder (Fig. 4): at $\Delta \approx 1$, FP4 training is stable and strong for 242 steps (train reward 0.80–0.85, eval 0.794 at step 220; run ended early by a data-loader file race, unrelated to training). At $\Delta=10$ it survives its full 24h (~ 420 steps), volatile in train reward but climbing in eval (0.802 at step 400). At $\Delta=32$ it collapses by step ~ 100 (train reward 0.38; eval already 0.17) and stays collapsed (0.303 at 400). The BF16 twin at the same lag collapsed at step ~ 170 from a higher peak (0.752 vs. 0.637): quantization and staleness *compound* — the boundary moves ~ 70 steps earlier and the pre-collapse peak drops — and at this horizon the $\beta=0$ boundary sits between $\Delta=10$ and 32. The rescue arms apply the *same* theory dose $\beta c = 0.0045$ used throughout, with no quantization-specific retuning. At $\Delta=10$ this is fully sufficient: healthy

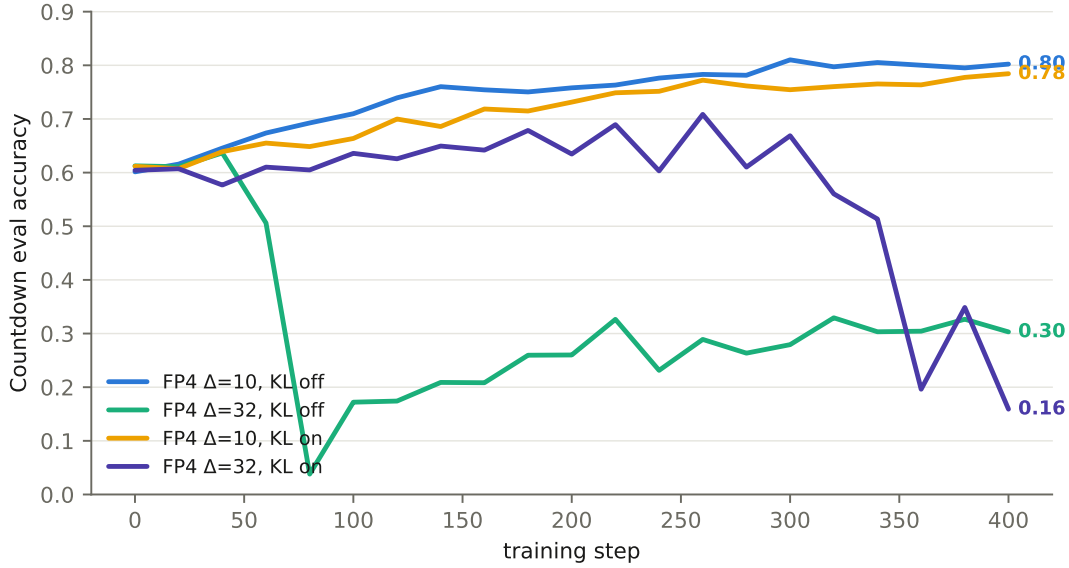


FIGURE 4. FP4 $\beta=0$ ladder and rescue arms. Blue/aqua: $\beta=0$ at $\Delta=10$ (stable) and $\Delta=32$ (collapses by step ~ 100). Yellow/violet: the same theory dose $\beta c = 0.0045$ at $\Delta=10$ (stable through 24 h) and $\Delta=32$ (holds $\sim 3\times$ past the $\beta=0$ boundary, degrades late).

through 24 h, eval 0.784 at step 400. At $\Delta=32$ it carries training through and far beyond the twin’s collapse window (0.636 at 100, 0.679 at 180, peak 0.709 at 260 — a $\sim 3\times$ extension of the boundary), but the climb is slower and noisier at reduced altitude, and late in the 24 h run evals degrade (0.51 at 340 $\rightarrow$ 0.16 at 400): the unscaled dose delays rather than fully prevents the compounded failure. A quantization-aware dose scaling (treating the quantization gap as extra effective staleness) is the natural refinement — untested.

Finding 6 (negative): the asymmetric arm fails informatively. The asymmetric arm (BF16 trainer, QDQ sampler, KL on at $\beta c = 0.0045$) degrades monotonically from the start: eval 0.56 at step 20, 0.04 at 100, 0.001 by step 180; cancelled at 200. Mechanism: the surrogate’s persistent quantization gap enters the advantage-level (score-function) penalty un-centered, acting as a global negative reward that suppresses all sampled behavior — not corrective QAT pressure. A differentiable KL-in-loss path, or centering of the gap, would be needed for this setting.

Finding 7: the original-hypothesis cell — staleness-weighted KL under FP4 heterogeneity. The batch that motivated this project: three arms, all symmetric NVFP4 QDQ with the heterogeneous offsets $\Delta_i \in \{1, 4, 10, 32\}$ at `async_level 32` — a $\beta=0$ control, a uniform dose $w=0.0045$ ($\beta=0.016$, global Δ -source), and per-rollout calibrated c_i ($\beta=0.005$) (Fig. 5). The control collapses at $\sim$ step 205 (train reward 0.75 at 140 $\rightarrow$ 0.51 at 200 $\rightarrow$ 0.26 at 212; eval 0.644 at 180 $\rightarrow$ 0.161 at 200) and never recovers through step 327 (eval 0.163 at 320; the run ended early by a data-loader file race, unrelated to training). The boundary at ~ 205 sits between the FP4-homogeneous $\Delta=32$ collapse (~ 100) and the BF16-heterogeneous collapse (~ 230), consistent with an intermediate effective mismatch. Both KL arms ran their full 24 h with zero errors and ended healthy (final steps $\sim 361/381$; last evals at step 360) — notably, under the heterogeneous mixture the uniform dose did *not* reproduce the late (~ 340) degradation of its homogeneous- $\Delta=32$ counterpart; the mixture’s milder mean lag presumably explains it. The verdict on staleness-weighting, from the eval curves: peaks 0.771 (per-rollout, step 360) vs. 0.764 (uniform, step 340), finals 0.771 vs. 0.764 at step 360 — both gaps under 0.01, a *tie* at one seed, with the lead trading at matched steps (0.705 vs. 0.711 at 200; 0.756 vs. 0.741 at 300; 0.742 vs. 0.746 at 320). (Train reward frequently favored the per-rollout arm — e.g. 0.807 vs. 0.735 at 200 — but that lead traded on batch noise; we judge on evals.) Either way, the confirmed part stands: staleness-weighted KL rescues FP4-heterogeneous training where $\beta=0$ collapses — the



FIGURE 5. The original-hypothesis cell: symmetric NVFP4 with heterogeneous staleness $\Delta_i \in \{1, 4, 10, 32\}$. Yellow ($\beta=0$) collapses at $\sim$ step 205 and never recovers; both KL arms — per-rollout staleness-weighted c_i (blue) and uniform $w=0.0045$ (aqua) — train healthily through their full 24h horizons and end within 0.007 eval of each other.

original hypothesis’s rescue claim holds; the comparison against uniform dosing is a tie at this seed count.

4. NEXT STEPS

Paired seeds (≥ 3) on calibrated vs. best-uniform; quantization-aware dose scaling for the compounded mismatch (the $\Delta=32$ rescue’s late degradation suggests the unscaled dose is insufficient there); a differentiable KL-in-loss path or gap-centered variant for the asymmetric setting; dataloader retry-on-corrupt-parquet (the file race that ended the async-1 arm); Δ_i floor test; a second task (MATH). Deferred items and operational cautions are recorded in EXPERIMENT_ROADMAP.md §7 and report_fp4/OPERATIONS_LOG.md.

APPENDIX A. CONFIGURATIONS

Common: model=Qwen/Qwen3-4B-Instruct-2507, LR=1e-6, sampled_k=train_k=16, steps=720, type=tb, conf=COUNTDOWN.toml, seq_length=4096, eval_steps=20, reference_mode=ema, ema_alpha=0.9, kl_centering=false, vllm_logprobs=true, batch 512, wandb fp4_KL, branch fp4_KL.

Arm	β	async	offsets	Δ -source
homog10_approxKL_b005_vllmLP	0.005	10	10,10,10,10	per_rollout
hetero_approxKL_b005_perRollout	0.005	32	1,4,10,32	per_rollout
hetero_approxKL_global_w0045	0.016	32	1,4,10,32	global
hetero_approxKL_global_w015	0.0533	32	1,4,10,32	global
hetero_b0_async32	0	32	1,4,10,32	—
bf16_b0_async32	0	32	— (legacy rule)	—
fp4sym_smoke (60 steps)	0.005	10	—	global
fp4sym_b0_async1	0	1	—	—
fp4sym_b0_async10	0	10	—	—
fp4sym_b0_async32	0	32	—	—
fp4sym_approxKL_b005_async10	0.005	10	—	global
fp4sym_approxKL_b0016_async32	0.016	32	—	global
fp4sym_approxKL_b005_async10 [†]	0.005	10	—	global
fp4hetero_b0_async32	0	32	1,4,10,32	—
fp4hetero_approxKL_global_w0045	0.016	32	1,4,10,32	global
fp4hetero_approxKL_b005_perRollout	0.005	32	1,4,10,32	per_rollout

KL penalty: $-\beta \sum_t c \text{ clamp}(\log T_n - \log T_{n-\Delta}, \pm 10)$ added to advantages (no centering); c per-token (c_i) or global per the table. FP4 arms add `rollout_quant=nvfp4` + `fake_quant=nvfp4` (252 patched linears; embeddings, `lm_head`, norms BF16). [†]BF16 trainer, QDQ sampler (asymmetric; `fake_quant` off).

APPENDIX B. EVAL TABLE (SELECTED STEPS)

Step	100	200	300	400	500	final
Homog. $\Delta=10$ reference	0.731	0.754	0.773	0.799	0.806	0.828 (680)
Hetero, per-rollout c_i	0.725	0.764	0.811	0.799	0.815	0.810 (540)
Hetero, uniform $w=0.0045$	0.722	0.759	0.782	0.794	0.817	0.806 (560)
Hetero, uniform $w=0.015$	0.670	0.730	0.763	0.774	0.801	0.788 (560)
Hetero, $\beta=0$	0.725	0.662	0.002	0.123	—	0.222 (480)
Homog. $\Delta=32$, $\beta=0$	0.752	0.140	0.246	0.271	0.319	0.340 (540)
FP4 $\Delta \approx 1$, $\beta=0$	0.726	0.786	—	—	—	0.779 (240)
FP4 $\Delta=10$, $\beta=0$	0.710	0.758	0.810	0.802	—	0.802 (400)
FP4 $\Delta=32$, $\beta=0$	0.172	0.260	0.279	0.303	—	0.303 (400)
FP4 $\Delta=10$, KL on	0.664	0.731	0.755	0.784	—	0.784 (400)
FP4 $\Delta=32$, KL on	0.636	0.635	0.669	0.159	—	0.159 (400)
FP4 asym. (BF16 trainer)	0.036	—	—	—	—	0.001 (180)
FP4-hetero, $\beta=0$	0.716	0.161	0.254	—	—	0.163 (320)
FP4-hetero, uniform $w=0.0045$	0.659	0.711	0.741	—	—	0.764 (360)
FP4-hetero, per-rollout c_i	0.649	0.705	0.756	—	—	0.771 (360)